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Targeting endolysosomal acidification inhibits poxvirus entry and replication

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Abstract

Background Orthopoxviruses, including monkeypox virus (MPXV) and cowpox virus (CPXV), pose significant public health threats, yet treatment options remain limited. Host-targeted antivirals could circumvent viral resistance by disrupting essential host pathways. Vacuolar H⁺-ATPase (V-ATPase), a proton pump critical for endolysosomal acidification, is implicated in poxviral entry and replication, making it a promising therapeutic target.

Methods We screened 268 antiviral compounds for activity against CPXV and identified Diphyllin, a V-ATPase inhibitor. In vitro assays (plaque reduction, time-of-addition, qPCR, TEM) evaluated its mechanism. V-ATPase dependency was confirmed using inhibitors (Bafilomycin A1, KM91104) and TMEM175 channel activators (DCPIB, arachidonic acid). Combination treatment assays in vitro were performed to evaluate the potential synergistic inhibitory effects of Diphyllin and Tecovirimat on viral infection. In vivo efficacy was assessed in CPXV-infected mice through analysis of viral titers, histopathology, cytokine levels, as well as survival experiments to evaluate the therapeutic benefit of Diphyllin alone and in combination with Tecovirimat.

Results Diphyllin potently inhibited CPXV and vaccinia virus (VACV) replication (EC₅₀ = 0.69 μM) without cytotoxicity. It blocked post-entry stages by disrupting V-ATPase-mediated endolysosomal acidification, suppressing viral uncoating and morphogenesis. Other V-ATPase inhibitors and TMEM175 activators similarly restricted infection, validating endolysosomal pH as a key vulnerability. In mice, Diphyllin reduced viral loads in spleen, liver, and adipose tissue, attenuated inflammation, and downregulated pro-inflammatory cytokines. Importantly, Diphyllin treatment improved the survival of CPXV-infected mice, demonstrating a measurable therapeutic benefit. Moreover, combination therapy with Tecovirimat further lowered the effective drug doses and yielded additional survival benefits, supporting the therapeutic potential of this ingredient.

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Conclusions Diphyllin is a broad-spectrum anti-poxviral agent targeting host endolysosomal acidification. Its efficacy in vitro and in vivo, coupled with synergistic effects of TMEM175 activation, highlights host pH modulation as a therapeutic strategy. These findings support further development of V-ATPase inhibitors for treating poxvirus infections.

Keywords Poxvirus, V-ATPase, Diphyllin, Endolysosome, Antiviral therapy

Background

Poxviruses such as cowpox virus (CPXV), monkeypox virus (MPXV), vaccinia virus (VACV), and variola virus (the causative agent of smallpox) are a family of large, enveloped viruses with linear double-stranded DNA genomes. Entry into host cells can occur either through fusion or by means of endocytosis with the host cell membrane, and this process involves at least 15 viral proteins [1]. Once inside the host cell, the virus initiates early gene transcription events and proceeds with DNA replication synthesis at specific cytoplasm sites known as “viral factories” [2, 3]. Since July 2022, the World Health Organization (WHO) has issued the highest level of alert due to the rapid spread of MPXV. It declared the MPXV outbreaks happening in multiple countries and regions around the world as an “international public health emergency of concern.” As of October 2023, the United States has reported a cumulative total of 30,861 confirmed cases and 54 deaths related to MPXV infection. While symptomatic treatment is common for most poxvirus infections, including MPXV, several antiviral agents developed for smallpox are now being used against MPXV. These include Brincidofovir, which works by inhibiting viral DNA replication; Tecovirimat, which inhibits the viral envelope protein to prevent cell-to-cell spread; and Cidofovir, which inhibits viral DNA polymerase. These drugs have shown promising antiviral activity and are employed as therapeutic options for MPXV [4–7]. However, recent randomized clinical trials have shown that although Tecovirimat is generally safe and well tolerated, it does not significantly improve clinical outcomes in human MPXV patients. In particular, a randomized controlled trial conducted in the Democratic Republic of Congo reported that Tecovirimat did not shorten the time to lesion resolution in clade I MPXV patients compared with placebo [8]. Similarly, interim analyses from the international STOMP trial for clade II MPXV also indicated no significant difference in outcomes between Tecovirimat-treated patients and placebo (Global Biodefense Staff [9]. Therefore, despite the availability of these antivirals, treatment options remain limited, and their real-world clinical effectiveness is still poorly defined [10–13]. Moreover, prolonged use of antiviral medications also brings the risk of viral immune escape [14]. In light of these limitations, recent studies have begun to explore combination antiviral strategies. Notably, Witwit et al. demonstrated that combining Tecovirimat with

host-targeted agents such as mycophenolate mofetil or the N-myrystoyltransferase inhibitor IMP-1088 produced potent synergistic anti-MPXV effects in vitro [15]. In addition, co-administration of tecovirimat with cidofovir was also reported to exert synergistic antiviral activity against orthopoxviruses and to improve outcomes in animal models [16]. Consequently, it is important to develop novel therapies that target host factors or virus-host interactions, as well as to find efficient combination therapies for treating of poxvirus infections.

Previous studies have provided evidence that the acidification of endosomes plays a crucial role in facilitating the fusion of vaccinia virus (VACV) with the host cell membrane, thereby promoting viral entry into the host cell [17]. More specifically, accumulating evidence indicates that vaccinia virus virions undergo low-pH triggered fusion predominantly within late endosomes, rather than mature lysosomes [18]. Alan et al. reported that VACV mature virions (MV) bound to the cell surface at low pH triggered the fusion between MV and the plasma membrane [19]. The Vacuolar ATPase (V-ATPase) is ubiquitously expressed in cells across diverse tissues and functions as a proton pump on the membranes of endosomes and lysosomes [20]. By persistently pumping hydrogen ions into endosomes and lysosomes, V-ATPase generates an acidic environment that is indispensable for the normal functioning of these organelles as well as other cellular processes [21]. Diphyllin is a recently identified inhibitor of V-ATPase, which has the capacity to raise the pH of endosomes and lysosomes [22]. It is a natural compound sourced from medicinal plants such as *Cleistthus collinus*, *Justicia gendarussa*, and *Haplophyllum bucharicum* [23–26]. There have been documented reports showing that Diphyllin can effectively impede the replication of Zika virus [27] and influenza virus [28]. Michal et al. discovered that Diphyllin exhibits broad-spectrum antiviral activity and is capable of suppressing the replication of a wide variety of viruses, including both enveloped RNA and DNA viruses [29]. Inhibition of V-ATPase results in rapid alkalization of endosomes and lysosomes. Recent studies have identified TMEM175 as an additional proton-efflux channel capable of modulating endolysosomal pH [30]. When TMEM175 activity is enhanced, it similarly promotes alkalization of these organelles. Consistent with this, arachidonic acid (ArA) and DCPIB, both known activators

of TMEM175, increase endolysosomal pH by stimulating proton efflux through this channel [31].

In this study, we screened a chemical compound library and identified the V-ATPase inhibitor Diphyllin as a potent inhibitor of poxviral replication. Consistent with this finding, other V-ATPase inhibitors, including Bafilomycin A, demonstrated similar antiviral activity. In line with this observation, the raising of endolysosomal alkalinity with chemical inducers of the proton channel TMEM175 also inhibited poxviral replication in cells. Moreover, Diphyllin alone or in combination with Tecovirimat reduced CPXV replication and virus-induced inflammation in mice. Collectively, these results suggest that endolysosomal acidification is a key step in the poxvirus lifecycle and that drugs targeting this process represent promising therapeutic candidates against poxvirus infections.

Methods

Cell culture

J2 virus-transformed macrophages (J2 macrophages), which are immortalized murine macrophages that are highly susceptible to CPXV *in vitro*, were generated in-house. These cells were cultured in Dulbecco's Modified Eagle Medium (DMEM, Pricella, Wuhan, CN) supplemented with 10% heated-inactivated fetal bovine serum (FBS) and 1% Penicillin-Streptomycin Solution at 37 °C in a humidified 5% CO₂ incubator. Vero (ATCC, Cat# CRL-1586) and BS-C-1 cells (ATCC, Cat# CCL-26) were cultured and maintained in DMEM and MEM, and they were used for viral amplification and viral titer determination. TMEM175 overexpressing HeLa cells were kindly provided by Dr. Hao-xin Xu. Bone-marrow derived macrophage (BMDM) was isolated from C57BL/6 mice infected with CPXV. Briefly, the femur and tibia were aseptically dissected, and the bone marrow was flushed into a sterile tube using cold RPMI-1640 medium (Pricella, Wuhan, CN). The marrow suspension was gently pipetted up and down with a 10 mL pipette to dissociate the tissue and then passed through a 70 µm cell strainer to remove debris and clumps. The resulting cell suspension was centrifuged at 1,500 rpm for 5 min, and the pellet was washed once with fresh RPMI-1640 medium. After the final centrifugation, cells were resuspended in complete growth medium (RPMI-1640 supplemented with 10% FBS, 1% penicillin-streptomycin, and 20 ng/mL recombinant murine GM-CSF) and plated in petri dishes. The cells were cultured for 7 days, with fresh medium added on day 3 or 4, to allow for differentiation into mature BMDMs. Human monocytic THP-1 cells (ATCC, Cat# HIB-202) were maintained in RPMI-1640 medium, supplemented with 10% heat-inactivated fetal bovine serum (FBS), 1% penicillin-streptomycin, and 0.05 mM β-mercaptoethanol. Cells were kept in a

humidified incubator at 37 °C with 5% CO₂ and passaged regularly to maintain logarithmic growth.

Late endosome and lysosome staining

Vero cells (5×10^5 cells /well) were cultured in 6-well plate overnight, incubating the cells for 2 h under growth condition with 1 µM LysoTracker Red (L12492, Thermo-Fisher). Replace the loading solution with fresh medium and observe the cells using a fluorescence microscopy fitted with the correct filter set.

Cathepsin B/D activity assay

Cathepsin B and D activities were measured using commercial assay kits (ab65300 and ab65302, Abcam, Cambridge, MA, USA) according to the manufacturer's protocols. In brief, Vero cells (1×10^4 cells /well) were cultured in 96-well plate overnight, infected cell with CPXV and add 50 µL of Cathepsin B Reaction Buffer to the sample and controls wells. Add 2 µL 10 mM Cathepsin B Substrate Ac-RR-AFC to all wells. Incubate at 37 °C for 1–2 h protected from light. The fluorescence was measured using a plate reader at excitation/emission wavelengths of 400/505 nm.

Virus strain

Cowpox virus (CPXV, Brighton Red strain), CPXV-GFP (Brighton Red strain), vaccinia virus (VACV, Western Reserve strain), VACV-GFP (Western Reserve strain), and myxoma virus (MYXV-GFP, Lausanne strain) were used in this study as previously described (Mohamed et al., 2009). Viruses were propagated in BS-C-1 cells by infection at a multiplicity of infection (MOI) of 0.05. After 72 h of incubation, cells were harvested and washed twice with 1× phosphate-buffered saline (PBS). Cell pellets were then resuspended in PBS containing 0.1% bovine serum albumin (BSA), followed by three cycles of freezing and thawing to release the viral particles.

Drug screening

A concentration of 2.5 µM was determined to be non-cytotoxic for all tested compounds and was therefore selected for subsequent antiviral screening. Vero cells were infected with cowpox virus (CPXV) at a multiplicity of infection (MOI) of 0.05 for 2 h at 37 °C. Following viral adsorption, the inoculum was removed, and the cells were incubated in culture medium containing 2.5% fetal bovine serum (FBS) and 1% Penicillin-Streptomycin Solution and 2.5 µM of the respective monomer compound. After 48 h of incubation, cells were fixed with 4% paraformaldehyde for 20 min at room temperature and stained with 0.1% crystal violet. After rinsing with water, viral plaques were visually counted, and the inhibition rate was calculated based on plaque reduction compared to the untreated control group.

EC₅₀ assay

Vero cells (1×10^4 cells/well) were seeded into 96-well plates and incubated overnight at 37 °C to allow adherence. Cells were then infected with cowpox virus (CPXV) at a multiplicity of infection (MOI) of 0.05 for 2 h. After viral adsorption, the medium was replaced with fresh medium containing varying concentrations of Diphyltin: 0.625, 1.25, 2.5, 5, 10, 20, 30, 40, 50, 80, and 100 µM. In parallel, Tecovirimat was evaluated under the same assay conditions using a concentration range of 0.0005–0.1 µM (0.0005, 0.001, 0.002, 0.005, 0.007, 0.009, 0.01, 0.02, 0.03, 0.04, 0.08, and 0.1 µM). Following treatment, Plates were incubated at 37 °C for 48 h. Following incubation, 100 µL of the CellTiter-Glo® Luminescent Cell Viability Assay reagent was added to each well. The plates were incubated at room temperature for 5 min to allow the luminescent reaction to develop. The resulting luminescence, proportional to ATP levels indicating metabolically active cells, was measured using a multimode microplate reader at 560 nm under appropriate excitation conditions.

Antibodies and reagents

Monoclonal antibodies (mAbs) against VACV proteins E3L (CSB-PA322729ZA01VAA), L1R (CSB-PA324949ZA01VAA), VACWR117 (CSB-PA356237XA01VAA), and D8L (CSB-PA322653LA-01VAA) were obtained from CUSABIO (Wuhan, China). ATP6V1A (#39517), LC3B(#3868), Phospho mTOR (#5536T), mTOR (#2983), and β-actin antibody (#4967) were purchased from Cell Signaling Technology (CST, Danvers, MA, USA). ATP6V0C (#ab230323) was purchased from Abcam (Cambridge, England). HIF-α (#20960-1-AP), P62 (#18420-1-AP) were obtained from Proteintech (Wuhan, China). A library of 268 antiviral monomeric compounds was procured from TargetMol (Boston, MA, USA). Diphyltin (TN1103) and Tecovirimat (T17026) were obtained from TargetMol. KM91104 (HY-135474), Bafilomycin A1 (BafA1, HY-100558), DCPIB (HY-103371), Cleistanthin B (HY-N9351), CID-1,067,700 (HY-13452) and Aloxistatin (HY-100229) were purchased from MedChemExpress (Shanghai, China).

Cytotoxicity assay

The cytotoxicity of Diphyltin, DCPIB and Tecovirimat was assessed through a Cell Counting Kit (CCK8) assay. Briefly, Vero cells were first seeded into 96-well plates and then cultured at 37 °C for 12 h. Subsequently, the medium was replaced with fresh medium containing Diphyltin and DCPIB that had been serially diluted. After that, the cells were incubated for an additional 48 h. Next, the culture medium was discarded, and 100 µL of CCK-8 solution (purchased from Beyotime, Shanghai, CN) was added to each well. Subsequently, the plates were incubated at 37 °C for 1 h. Cell viability was determined by

measuring the absorbance at 450 nm using a microplate reader (Thermo Fisher Scientific, MA, USA). The mean optical density (OD) values obtained from six wells for each treatment were adopted as the cell viability index.

Western blot

Vero cells and BS-C-1 cells were seeded at a density of 5×10^5 cells per well in 6-well plate. Cell lysates were prepared using SDS lysis buffer (comprising pH7.5 1 M Tris-HCl, 10% SDS, ddH₂O) supplemented with 1X Complete protease inhibitor cocktail (from Solarbio) and 1X phosphatase inhibitor cocktail (from Solarbio). Insoluble material was removed by centrifugation at 12,000 rpm for 15 min at 4 °C. Subsequently, the soluble lysates were transferred to new Eppendorf tubes. Protein concentrations were determined through the BCA Protein Assay (manufactured by ThermoFisher). The antibodies utilized for western blot in this study have been described in the “Antibodies and reagents”. Chemiluminescence signals were detected by means of the ChemiDoc MP System from Bio-Rad.

Mouse infection and viral titer determination

Mice were purchased from the Experimental Animal Center of the College of Medicine, Xi'an Jiaotong University (Xi'an, China). Male and female C57BL/6 mice, aged 8–10 weeks, were intraperitoneally infected with 1×10^6 pfu of wild type CPXV - GFP. At 3.5 days post-infection, the spleen, visceral fat pad, and liver were harvested using a tissue disruptor in DMEM containing 2.5% fetal calf serum and 1% Penicillin-Streptomycin Solution. Serial 10-fold dilutions of the tissue extracts were made, and the viral titer was determined through plaque assays on Vero cells. After 2 days of infection, the Vero cells were fixed with 5% formaldehyde for 20 min and then stained with 0.1% crystal violet. After rinsing with water, the plaques were counted visually. For the survival experiments, C57BL/6 mice (6–8 weeks old) were maintained under SPF conditions. All mice were intraperitoneally infected with 5×10^6 PFU of cowpox virus (CPXV) suspended in 100 µL of sterile PBS. After infection, mice were randomly assigned to treatment or control groups ($n = 10$ per group). Mice in the treatment group received Diphyltin at 5 mg/kg body weight via intraperitoneal injection once daily for the first five days post-infection, with each dose prepared in a total volume of 100 µL PBS. Control mice received an equal volume of PBS on the same schedule. All injections were performed using sterile syringes. For the Tecovirimat treatment experiments, mice were administered the drug by oral gavage. A 1-mL sterile syringe fitted with a plastic gavage needle was gently inserted into the esophagus until it reached the stomach. Each mouse received 50 µL of Tecovirimat solution per dose. Tecovirimat was administered once daily for

the first five days post-infection. Control group animals received an equal volume (50 μ L) of sterile saline via the same gavage procedure. All gavage administrations were performed carefully to minimize distress and ensure accurate delivery of the drug into the stomach.

Following infection and treatment, survival, clinical symptoms, and body weight were monitored daily for 9 days. Mice showing severe distress or more than 20% body weight loss were humanely euthanized according to institutional ethical guidelines.

Real time-PCR

TRIZOL reagent (ABclonal, Wuhan, CN) was used to extract total RNA from 20 mg visceral fat pad, and liver according to the instructions provided by the reagent manufacturer. Subsequently, cDNA was synthesized using the ABScript Reverse Transcription Master Mix (from ABclonal). Real time-PCR was carried out with the 2X Universal SYBR Green Fast qPCR mix (from ABclonal) on the Bio-Rad CFX Connect Real-Time PCR Detection System. Gene expression was quantified using the comparative Ct ($2^{(-\Delta\Delta Ct)}$) method. The Ct value for each target gene was normalized to the geometric mean of the Ct values for *ACTB* to calculate ΔCt . The $\Delta\Delta Ct$ was derived by subtracting the mean ΔCt of the control group from the ΔCt of each sample. Fold-change in gene expression relative to the control group was expressed as $2^{(-\Delta\Delta Ct)}$. The primers utilized for the qPCR reactions are listed in the supplementary Table 1.

Histology

The visceral fat pad was taken out from the peritoneal cavity and then fixed in 10% Neutral Buffered Formalin (from Leica Biosystems). Subsequently, it was embedded in paraffin and sliced into 5 μ m sections. The tissue sections were stained with Hematoxylin and Eosin. The stained sections were imaged using a Leica DMD108 microscope.

Transmission Electron Microscopy (TEM)

Vero cells (5×10^5 cells/well) were cultured overnight in 6-well plate and subsequently infected with CPXV-GFP (MOI 0.05) for 2 h. Infection was carried out in the presence of absence of Diphyltin. Following infection, Samples were collected and fixed with a solution containing 2.5% glutaraldehyde. Cells were then washed in 0.1 M sodium cacodylate buffer and treated with 0.1% Millipore-filtered cacodylate buffered tannic acid, post-fixed with 1% buffered osmium and stained en bloc with 0.1% Millipore-filtered uranyl acetate. The samples were dehydrated in increasing concentrations of ethanol and then infiltrated and embedded in LX-112 medium. The samples were then polymerized in a 60 $^{\circ}$ C oven for approximately three days. Ultrathin sections were cut using a

Leica Ultracut microtome (Leica, Deerfield, IL) and then stained with uranyl acetate and lead citrate in a Leica EM Stainer. The stained samples were examined in a JEM 1010 transmission electron microscope (JEOL USA, Inc, Peabody, MA) using an accelerating voltage of 80 kV. Digital images were obtained using an AMT imaging system (Advanced Microscopy Techniques Corp, Danvers, MA).

SiRNA transfection

Vero cells were seeded at a density of 2.5×10^5 cells per well in 12-well plates and transfected with 50 nM of small interfering RNA (siRNA) using 6 μ L of Lipofectamine RNAiMAX reagent (Invitrogen, USA), according to the manufacturer's instructions. Twenty-four hours post-transfection, the cells were infected with CPXV-GFP at a multiplicity of infection (MOI) of 0.05. Twelve hours post-infection, viral replication was assessed by measuring GFP fluorescence intensity, and viral plaques were subsequently stained for quantification. The specific siRNA sequences targeting ATP6V1A and ATP6V0A2 are listed below:

For siATP6V1A

- Forward sequence:
5'-CCUACGGGUUGGUAGUCAUAUTT-3'.
- Reverse sequence:
5'-AUAUGACUACCAACCCGUAGGTT-3'.

For siATP6V0A2

- Forward sequence: 5'-
GTTTCATTGCTGGGAAGCCAAGA-3'.
- Reverse sequence: 5'-
TTGGTGCCTGCACCAACGTAGA-3'.

RNA-seq analysis

The RNA-seq datasets were retrieved from the Gene Expression Omnibus (GEO), with the accession number being GSE120616. Subsequently, the data were integrated and subjected to analysis using R (version 4.4.0) along with the R package affy (version 1.82.0). The visualization of the data was carried out based on TOMicsVis (version 2.0.0) and ggplot2 (version 3.5.1). Moreover, the Gene set variation analysis (GSVA) and single sample gene set enrichment analysis (ssGSEA) were conducted based on the R package GSVA (version 1.53.2). Statistical analysis was performed by means of the "t-test" provided within TOMicsVis (version 2.0.0).

AlphaFold3-based structure prediction and analysis

Full-length human sequences of ATP6V0A2 (Primary Accession Numbers = Q9Y487), ATP6V0C (P27449),

and ATP6V1A (P38606) were obtained from UniProt. The small-molecule ligand diphyllin was represented by the SMILES: COC1=C(C=C2C(=C1)C(=C3C(=C2O)COC3=O)C4=CC5=C(C=C4)OCO5)OC. We used local-installed AlphaFold 3 (AF3) [32] with its extended diffusion-based architecture to predict the joint structure of the protein complex and the bound ligand. AF3 was executed in a GPU-accelerated conda environment, with standard databases mounted. The generated structure with highest ranking score was used for downstream analysis. PLIP (<https://plip-tool.biotec.tu-dresden.de/plip-web/plip/index>) was used for profiling of non-covalent interactions in the AlphaFold-predicted complex, delivering publication-ready figures and machine-readable data for detailed interface characterization.

Code availability

This study did not generate new original codes. AlphaFold3 predictions were produced using the official local AlphaFold3 pipeline (<https://github.com/Model3DBio/AlphaFold3-Conda-Install>):

```
python run_alphafold.py \
  • --db_dir /data/AlphaFold3/alphafold3/public_databases/ \
  • --model_dir /data/AlphaFold3/alphafold3/models/ \
  • --input_dir /data/Protein/VATP_af3_jsons \
  • --output_dir /data/Protein/VATP_af3_jsons \
  • --run_data_pipeline true \
  • --run_inference true.
```

The input json files were like {"name": "", "version": 2, "dialect": "alphafold3", "modelSeeds": [1], "sequences": [{"protein": {"id": ["A"], "sequence": ""}, {"ligand": {"id": ["B"], "smiles": "COC1=C(C=C2C(=C1)C(=C3C(=C2O)COC3=O)C4=CC5=C(C=C4)OCO5)OC"}]}], and input sequence were directly obtained from Uniprot (<https://www.uniprot.org/>).

Statistical analysis

Statistical analyses were conducted using Prism 8.0 (GraphPad). Line and bar graphs display the mean $\pm$ SEM for each data point. For comparisons involving more than two groups, one-way ANOVA followed by Tukey's or Bonferroni's post hoc tests was performed as appropriate. Non-paired two-tailed Student's *t* tests (with a 95% confidence level) were employed for comparing two groups. The variable "n" represents the number of animals utilized in each group. The specific statistical details can be found in the figure legends. *P* values lower than 0.05 were regarded as significant (**p* < 0.05, ***p* < 0.01, ****p* < 0.001). The Kaplan-Meier survival curve was plotted by means of Prism 8.0.

Results

Diphyllin inhibits poxvirus replication

A screening of 268 antiviral monomeric compounds from TargetMol identified Diphyllin as the most potent inhibitor of cowpox virus (CPXV), as shown in Fig. 1A-B. To validate this, Vero cells were infected with CPXV at a multiplicity of infection (MOI) of 0.05 and treated with various concentrations of Diphyllin. At 2.5 μ M, Diphyllin nearly completely blocked CPXV and VACV infection (Fig. 1C). The half-maximal effective concentration (EC_{50}) was calculated to be 0.69 μ M (Fig. S1A). Cell viability assays (Fig. S1A) confirmed that 2.5 μ M Diphyllin was non-cytotoxic. Further validation using GFP-expressing poxviruses demonstrated a dose-dependent reduction in viral fluorescence intensity following infection with CPXV, VACV, and myxoma virus (MYXV) (Fig. 1D), supporting the broad anti-poxviridae activity of Diphyllin. To assess its prophylactic potential, cells were pre-treated with Diphyllin 24 h before infection, and it was removed immediately prior to viral exposure. This pre-treatment still conferred measurable protection (Fig. S1B), indicating a preventive role. Given the limitations of Vero cells, which lack interferon responses, additional experiments were performed in THP-1 cells, J2 macrophages and bone marrow derived macrophages. Diphyllin maintained comparable antiviral efficacy (Fig. S1C). We also tested Cleistanthin B (Diphyllin O-glucoside), a structural analog of Diphyllin, which showed even stronger antiviral activity (Fig. S1D), potentially due to enhanced cellular uptake mediated by its glucoside moieties. To investigate the impact of Diphyllin on viral protein expression, we measured levels of four VACV proteins (E3L, VACWR117, L1R, D8L), which represents early, intermediate, and late gene expression stages. Diphyllin significantly suppressed expression of all four proteins, with the most pronounced reduction observed for L1R (Fig. 1E F). As L1R, a myristoylated membrane protein located on the intracellular mature virion (IMV), is essential for core penetration during the low pH-dependent membrane fusion process [33], the pronounced suppression of L1R expression by Diphyllin strongly suggests that Diphyllin impairs poxvirus-host cell membrane fusion and subsequent viral entry. A dose-response study confirmed that increasing Diphyllin concentrations led to progressively lower viral protein expression (Fig. S1E, S1F). qPCR analysis corroborated that Diphyllin inhibits acidification-dependent viral entry and subsequent early, intermediate, and late gene expression (Fig. S1G). To more specifically determine whether Diphyllin also affects viral genome replication and transcription, we quantified the expression of key poxviral genes involved in these processes. Diphyllin treatment significantly reduced the mRNA levels of replication-associated genes (A20R, D4R, E9L) and transcription-related genes (D1R,

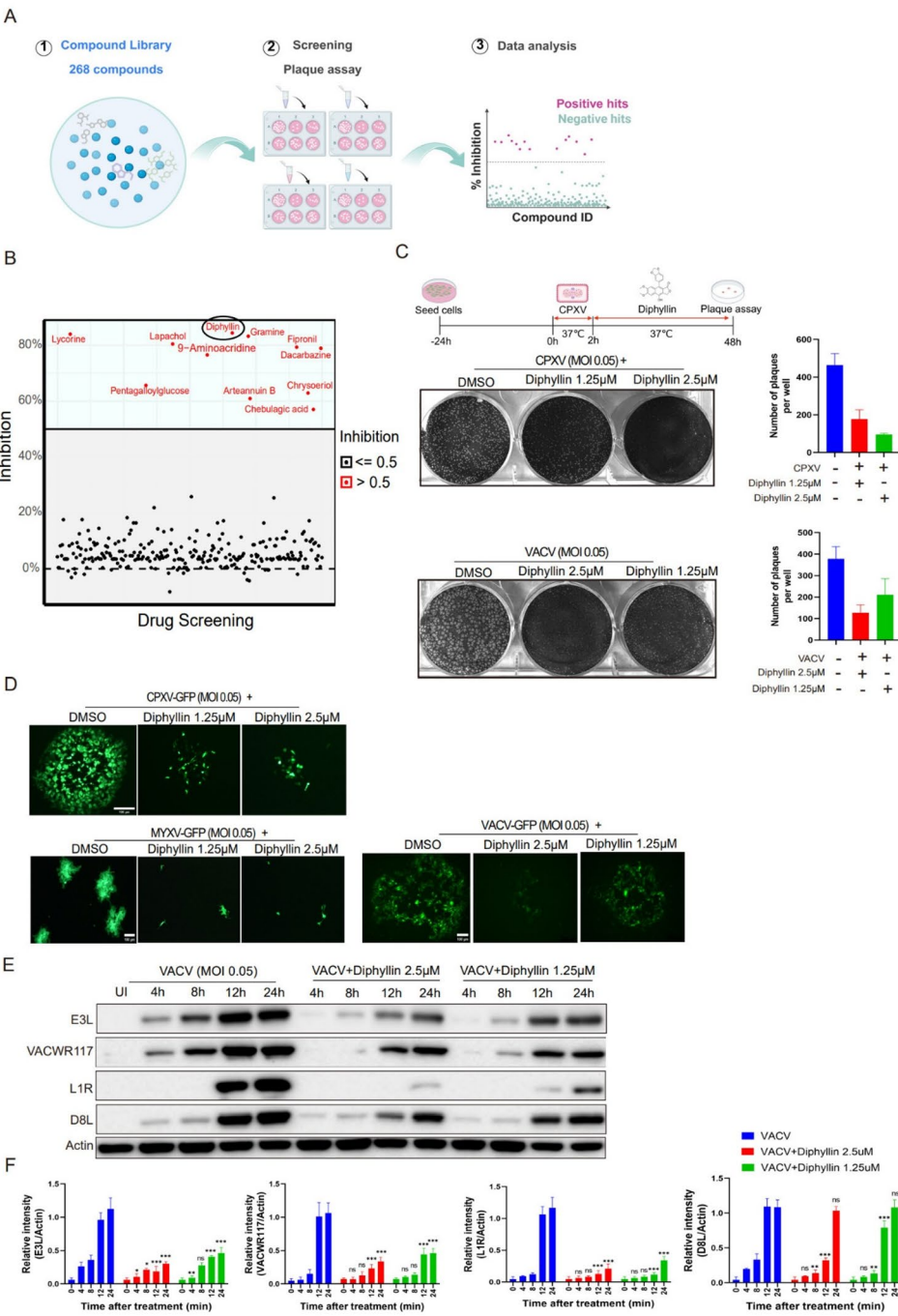


Fig. 1 Diphyllin inhibits poxvirus replication. **(A)** A library of 268 compounds was individually tested at 2.5 μM on Vero cells infected with cowpox virus (CPXV, MOI=0.05). After incubation, plaque assays were performed to determine antiviral efficacy by calculating the plaque reduction rate relative to untreated controls. **(B)** High-throughput screening of 268 antiviral monomeric compounds from the TargetMol library. **(C)** Experimental schematic illustrating the infection and drug treatment protocol used for subsequent viral assays unless otherwise specified. Plaque assay results showing CPXV-infected (MOI=0.05) Vero cell and VACV-infected (MOI=0.05) BS-C-1 cells treated with Diphyllin at 1.25 μM and 2.5 μM, compared to untreated infected cells. **(D)** GFP fluorescence imaging of Vero cells infected with CPXV-GFP, VACV-GFP, or MYXV-GFP (MOI=0.05), treated with Diphyllin at 1.25 μM and 2.5 μM, compared to untreated infected cells. **(E)** Western blotting analysis of VACV-infected BS-C-1 cells (MOI=0.05) treated with Diphyllin at 1.25 μM and 2.5 μM, compared to untreated infected cells. Protein levels of viral proteins E3L, WR117, L1R, and DBL were detected. **(F)** Quantification of Western blotting band intensities from three independent experiments presented as bar graphs. The mean value was calculated by one-way ANOVA with Tukey. (mean ± SD, $n=3$). * $P<0.05$, ** $P<0.01$, *** $P<0.001$

J3R, D6R) (Figure S1H). These results demonstrate that Diphyllin interferes with multiple essential stages of the poxviral life cycle, including viral entry, DNA replication, and mRNA transcription. In summary, Diphyllin exhibits strong antiviral activity against members of the Poxviridae family and represents a promising candidate for the development of host-targeted antiviral therapies.

Diphyllin inhibits poxvirus replication during intracellular biosynthesis and virus release stages

To determine the specific stages of the viral life cycle targeted by Diphyllin, we employed four treatment regimens (Fig. 2A–B) to assess antiviral effects based on the timing of drug administration. At 4 °C, cellular and viral metabolic activities are largely arrested, permitting viral adsorption but preventing internalization. In group #1, pretreating cells with Diphyllin for 1 h did not inhibit viral adsorption (Fig. 2C–E). Similarly, in group #2, co-incubation of virus and Diphyllin at 37 °C for 1 h did not block viral entry, indicating that Diphyllin does not inactivate the virus directly. These results suggest that Diphyllin likely acts via host cell pathways to inhibit infection. In contrast, groups #3 and #4, where Diphyllin was added post-attachment, exhibited marked reductions in infection, indicating that Diphyllin inhibits stages following viral binding to the cell surface. This pattern of inhibition was consistent across CPXV, VACV, and MYXV infections (Fig. 2C–E). To further verify the antiviral action of Diphyllin at the intracellular replication and virus release stages, transmission electron microscopy (TEM) was performed. As shown in Fig. 2F, Diphyllin treatment disrupted viral morphogenesis and release, confirming inhibition at these later stages. To refine the temporal window of Diphyllin's efficacy, we added the drug at various time points post-infection and quantified plaque formation after 48 h. As shown in Fig. S2A–C, maximal suppression of viral replication by Diphyllin was observed when the compound was administered within the first 4 h post-infection, a period during which replication was nearly abolished. In summary, Diphyllin effectively inhibits poxvirus replication by targeting the intracellular biosynthesis and virion release stages of the viral life cycle.

Diphyllin suppresses poxvirus replication by targeting V-ATPase

Having identified that Diphyllin inhibits poxvirus replication at the intracellular replication and release stages, we next aimed to elucidate the underlying cellular mechanism. Previous studies have suggested that Diphyllin functions as a V-ATPase inhibitor, however, its specific molecular targets remain insufficiently characterized. Notably, the structurally similar derivative PHY34 was shown to directly target the V-ATPase [34]. Given that

autophagic flux relies on endolysosomal acidification and degradative capacity, this finding further supports a model in which Diphyllin perturbs endolysosomal function by targeting the V-ATPase. To explore the specificity of Diphyllin binding within the V-ATPase complex, we used AlphaFold 3 (AF3) to generate structural predictions of each subunit in complex with Diphyllin. For each target, five independent stochastic predictions were performed. The resulting models were ranked by interface predicted TM-score (ipTM), and the top-ranked models were subjected to PLIP-based contact profiling and manual inspection (Fig. 3A and S3A). We selected three subunits, including ATP6V0A2, ATP6V0C, and ATP6V1A, for presentation. These subunits were chosen based on a combination of predicted drug ability, functional relevance to proton transport, and prior evidence of inhibitor binding: ATP6V0A2 for its known interaction with Diphyllin analogs [34], and ATP6V1A as the principal catalytic subunit of the V1 domain [35]. Among the subunits, ATP6V1 components, except for ATP6V1G2, exhibited higher average ipTM scores than ATP6V0 components. Notably, ATP6V0A2 displayed a high-confidence binding mode in which Diphyllin bound to its N-terminal domain. This domain acts as a cytosolic regulatory scaffold that mediates V₁–V₀ coupling, complex assembly and activity regulation, but does not directly participating in proton translocation [36]. Taken together, the AF3 models suggest a binding preference involving both ATP6V0A2 and ATP6V1 subunits. These findings extend previous observations that Diphyllin primarily inhibits V-ATPase proton pump activity mediated by the V₀ domain, while also raising the possibility that Diphyllin may indirectly impair V₀ function through modulation of ATP hydrolysis by ATP6V1. This interpretation is supported by our functional data showing that siRNA-mediated knockdown of ATP6V1A effectively suppresses poxvirus infection (Fig. 4E).

To further validate the role of V-ATPase inhibition in suppressing poxvirus replication, we evaluated the antiviral activity of two well-characterized V-ATPase inhibitors-Bafilomycin A1 (BafA1) and KM91104. Both compounds significantly reduced viral plaque formation and GFP fluorescence intensity in CPXV-infected cells (Fig. 3B–D), indicating effective inhibition of viral replication. Similar antiviral effects were observed against VACV and MYXV (Fig. S3B–C). The half-maximal effective concentrations (EC₅₀) for BafA1 and KM91104 were determined to be 0.09 μM and 7.72 μM, respectively (Fig. S3D–E). Collectively, these findings support the conclusion that Diphyllin exerts broad-spectrum antiviral activity against Poxviridae viruses primarily through inhibition of V-ATPase function.

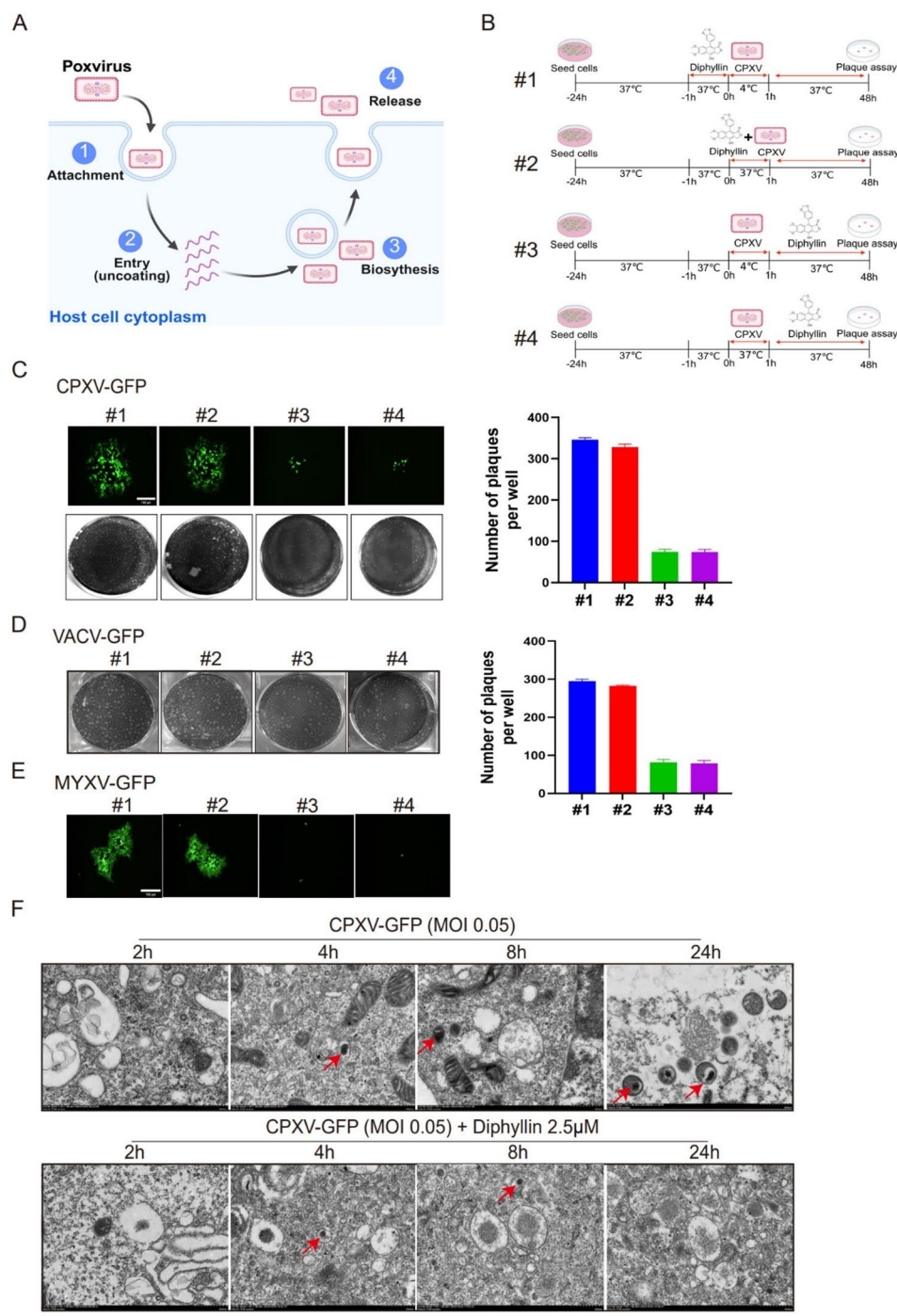


Fig. 2 Diphyllin inhibits poxvirus replication during intracellular replication and virus release stages. **(A)** Pattern diagram of the four stages of viral adsorption, entry, biosynthesis and release of poxviruses when they entered host cells. **(B)** Schematic illustration of four Diphyllin treatment regimens designed to identify the targeted stage(s) of the poxvirus life cycle. **(C)** The four treatment regimens were applied to cells infected with CPXV-GFP. Viral infection was assessed by plaque assay and by GFP fluorescence imaging. **(D–E)** The four treatment regimens were applied to cells infected with vaccinia virus (VACV-GFP) and myxoma virus (MYXV-GFP). Viral infection was assessed by plaque assay for VACV and by GFP fluorescence imaging for MYXV. **(F)** Vero cells were infected with CPXV-GFP for 2 h in the presence or absence of Diphyllin. Following infection, samples were collected and fixed at the indicated time points (2, 4, 8, and 24 h post-infection) for ultrastructural observation. Red arrows highlight arrested, non-progressing immature viral structures

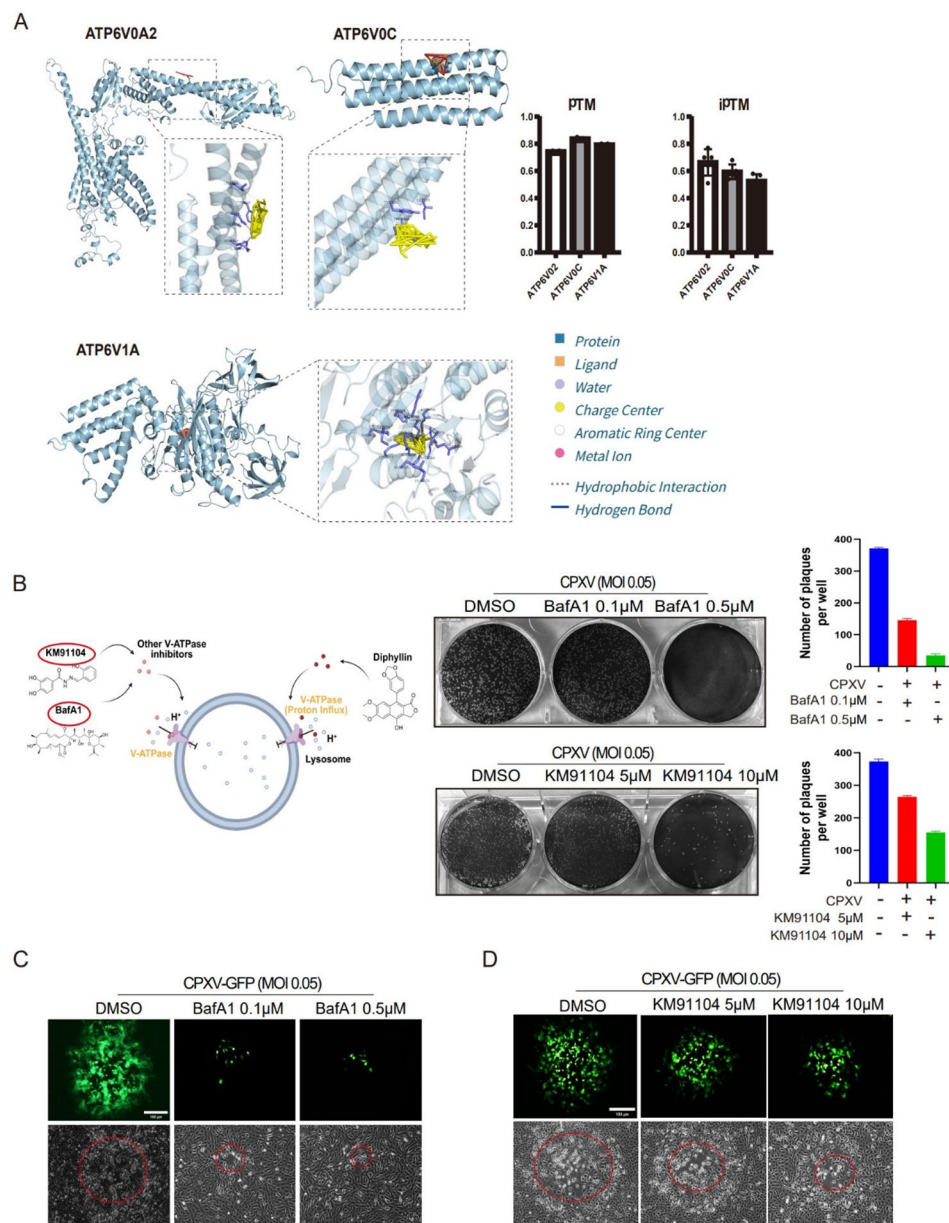


Fig. 3 Diphyllin suppresses poxvirus replication via V-ATPase inhibition. **(A)** Predicted binding modes of Diphyllin with human V-ATPase subunits. Cartoon views show the full-length monomeric predictions for the membrane-embedded V_0 subunits ATP6V0A2 (top left) and ATP6V0C (top right) together with the cytosolic V_1 catalytic subunit ATP6V1A (bottom left). **(B)** Plaque assay results of CPXV-infected Vero cells (MOI = 0.05) treated with BafA1 (0.1 μ M, 0.5 μ M) or KM91104 (2.5 μ M, 5 μ M), compared with untreated CPXV-infected control cells. **(C, D)** GFP fluorescence imaging of CPXV-infected cells (MOI = 0.05) treated with BafA1 and KM91104. Both V-ATPase inhibitors demonstrated dose-dependent suppression of viral replication

Endolysosomal acidification via V-ATPase is essential for poxvirus replication

V-ATPase, a proton pump located on the membranes of endosomes, plays a crucial role in maintaining endolysosomal acidity by translocating protons into the lumen [37]. Our results demonstrate that Diphyllin significantly reduces the expression of the viral protein L1R, which is essential for membrane fusion under acidic conditions, suggesting that Diphyllin may disrupt endolysosomal acidification. To investigate changes in

endolysosomal pH during poxvirus infection, we measured endolysosomal acidity in CPXV-GFP infected Vero cells at 24 h post-infection using LysoTracker staining. The results revealed a significant increase in endolysosomal acidification following infection, likely facilitating viral uncoating and entry (Fig. 4A-B). This finding was further supported by Lysosensor Green staining, which confirmed that CPXV infection enhances endolysosomal acidification, a process that was effectively inhibited by Diphyllin (Fig. S4A). Consistent with this, we observed

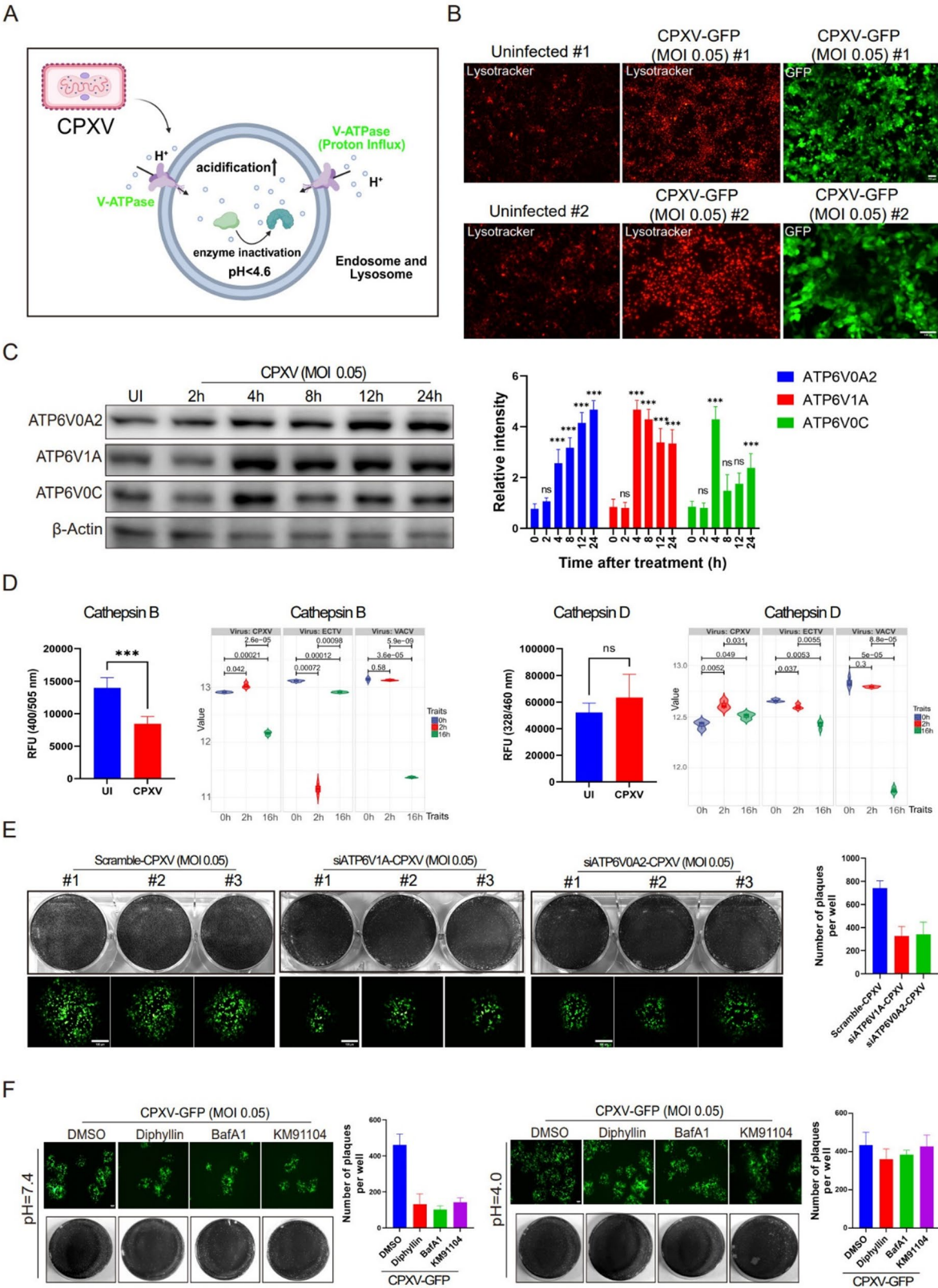


Fig. 4 (See legend on next page.)

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Fig. 4 Endolysosomal acidification via V-ATPase is essential for poxvirus replication. **(A)** Schematic diagram of CPXV regulating the acidic environment of endolysosomes. **(B)** Representative images of LysoTracker Red staining in Vero cells (left), LysoTracker Red staining in Vero cells infected with CPXV (MOI = 0.05) for 24 h (middle), and corresponding GFP fluorescence images of CPXV-infected cells at 24 h (right). **(C)** Western blotting analysis of V-ATPase subunits (ATP6V1A, ATP6V0A2, and ATP6V0C) in CPXV-infected Vero cells (MOI = 0.05) at different time points post-infection. The mean value was calculated by one-way ANOVA with Tukey. (mean $\pm$ SD, $n = 3$). * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$. **(D)** Quantification of Cathepsin B and D activity and gene expression in Vero cells. Bar plots show enzymatic activity of Cathepsin B and D measured using fluorogenic substrates. Violin plots represent transcriptomic expression levels of CTSB and CTSD genes in Vero cells following infection with CPXV, ECTV, or VACV for 0 h (blue), 2 h (red) and 16 h (green). UI: uninfected; CPXV: cowpox virus infected. Numerical values above the plots indicate two-tailed t-test P values. **(E)** Plaque assay and GFP fluorescence imaging results of CPXV (MOI = 0.05) infection in Vero cells after siRNA-mediated knockdown of ATP6V1A (middle panels) and ATP6V0A2 (right panels), compared with control siRNA-treated cells (left panels). **(F)** Plaque assay and GFP fluorescence imaging results of CPXV-infected Vero cells treated with Diphyllin, Bafilomycin A1 (BafA1), or KM91104 for 48 h under neutral (pH 7.4) and acidic (pH 4.0) conditions

a time-dependent upregulation of V-ATPase subunits during infection, suggesting enhanced V-ATPase activity (Fig. 4C). Interestingly, Cathepsin B activity was reduced 24 h post-infection, which is in agreement with transcriptomic data (Fig. 4D). This reduction may result from excessive acidification impairing protease function, thereby creating an environment that could favor viral egress. However, other hydrolytic enzymes, including Cathepsin D, K, G, and S, showed no significant changes (Fig. 4D and Fig. S4B). To further evaluate the role of V-ATPase, we used siRNA to knock down ATP6V1A and ATP6V0A2 in Vero cells (Fig. S4C), followed by infection with CPXV-GFP (MOI 0.05). Silencing these genes partially inhibited viral replication (Fig. 4E), further supporting the critical role of V-ATPase in promoting infection. Finally, to assess the importance of acidic conditions during viral entry, we treated cells with acidic PBS during CPXV adsorption and subsequently cultured them in media containing V-ATPase inhibitors. Under these conditions, the antiviral effect of the inhibitors was completely abolished (Fig. 4F, S4D), confirming that endolysosomal acidification is essential for poxvirus entry and replication. To distinguish whether late endosomes or lysosomes are essential for poxvirus entry, we used inhibitors to block lysosomal fusion (CID-1067700) and enzyme activity (Aloxistatin). Neither inhibitor reduced infection, indicating that viral replication relies on acidic endosomes rather than functional lysosomes (Figure S4E). Consistent with impaired endosomal acidification, Diphyllin increased HIF-1 α and blocked autophagic flux (accumulated LC3-II/p62) in infected cells by suppressing mTORC1 (reduced p-mTOR). These effects align with endolysosomal dysfunction and mTORC1 dysregulation and are consistent with Diphyllin disrupting pH-dependent poxviral entry (Figure S4F). However, the contribution of the integrated stress response (ISR) downstream of V-ATPase inhibition to Diphyllin's antiviral effects remain unclear and requires further investigation. Collectively, our results suggest that CPXV entry occurs in acidified endolysosomal compartments rather than in fully matured lysosomes. We therefore prefer the term endolysosomal to encompass these acidified vesicles (late endosomes/early lysosomes) that support viral fusion.

While previous studies have emphasized that many enveloped viruses “exploit the endolysosomal system to enter host cells” [38], our data now substantiate this model specifically for CPXV.

TMEM175-mediated endolysosomal alkalization restricts poxvirus replication

TMEM175 has recently been identified as an endolysosomal ion channel that facilitates proton efflux and works in conjunction with V-ATPase to maintain endolysosomal pH homeostasis (Fig. 5A). We hypothesized that the activation of TMEM175 could disrupt endolysosomal acidification, thereby impairing poxvirus replication. To test this hypothesis, we generated a TMEM175-overexpressing cell line (Fig. S5A), which displayed significantly reduced viral replication (Fig. 5B), suggesting that TMEM175 activation, and the resulting endolysosomal alkalization, impairs poxvirus infection. To further validate these results, we evaluated the antiviral effects of DCPIB, a well-established TMEM175 activator, which has a CC_{50} value of 45.55 μ M and an EC_{50} value of 7.122 μ M (Fig. S5B–C). Treatment with DCPIB resulted in a significant reduction in both viral plaque formation and CPXV-GFP fluorescence intensity, as shown in Fig. 5C. Similarly, we tested another TMEM175 activator, arachidonic acid (ArA), which also led to a reduction in viral plaque formation and CPXV-GFP fluorescence intensity (Fig. 5D). This inhibitory effect was similarly observed in VACV upon treatment with both DCPIB and ArA (Fig. S5D–E). Moreover, treatment with DCPIB markedly suppressed VACV mRNA and protein expression at early, intermediate, and late stages of replication. Notably, the vaccinia D8L protein was almost completely abolished. (Fig. 5E–F). Collectively, these results underscore the critical role of TMEM175 in regulating endolysosomal pH and suggest its potential as an antiviral target for inhibiting Poxviridae replication.

Diphyllin inhibits poxviral replication in vivo

To assess the antiviral efficacy of Diphyllin in vivo, we intraperitoneally co-administered CPXV (1×10^6 PFU) with Diphyllin (5 mg/kg) into 8–12-week-old mice ($n = 8$ per group). At 3.5 days post-infection, the fat, liver,

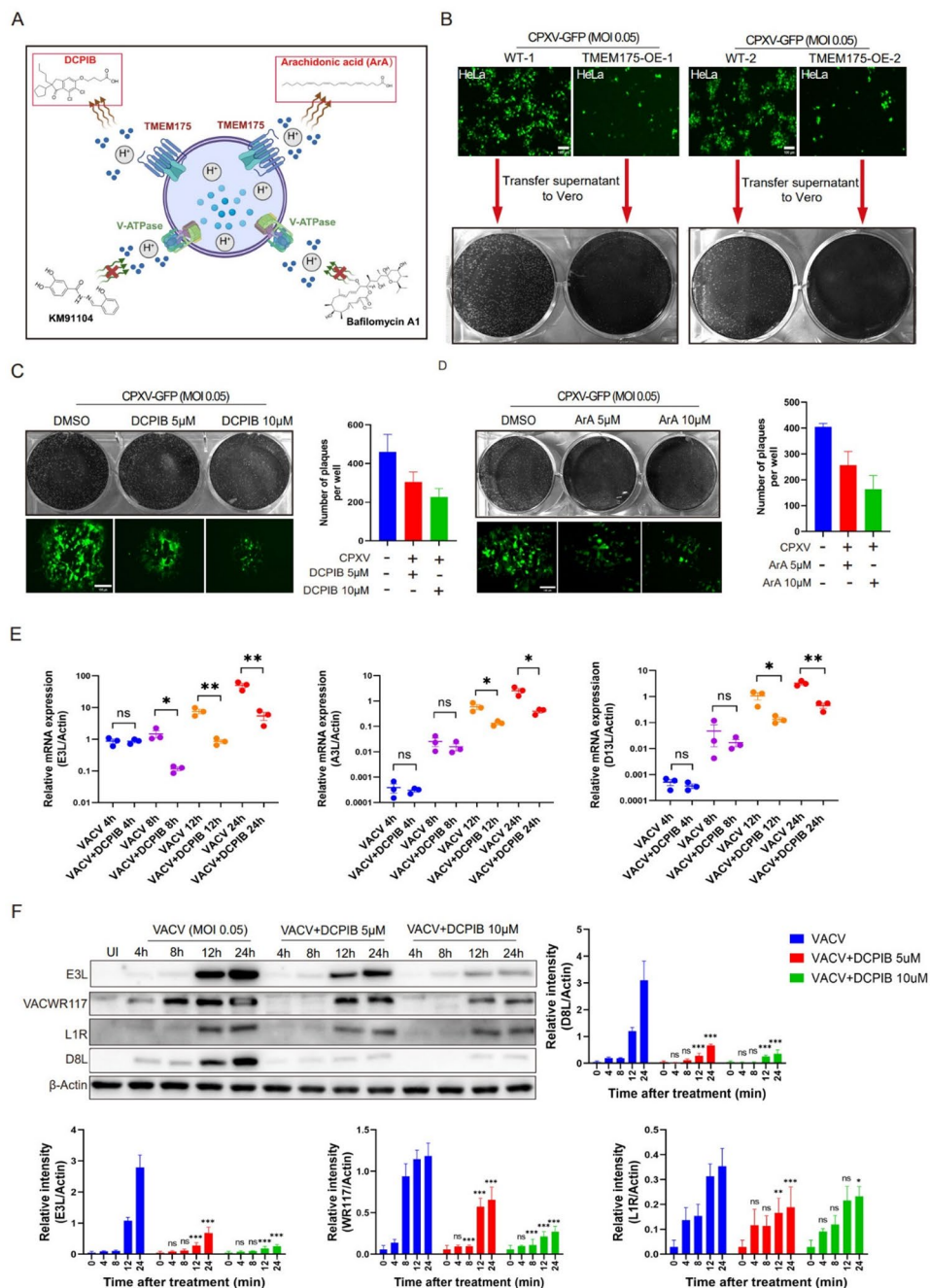
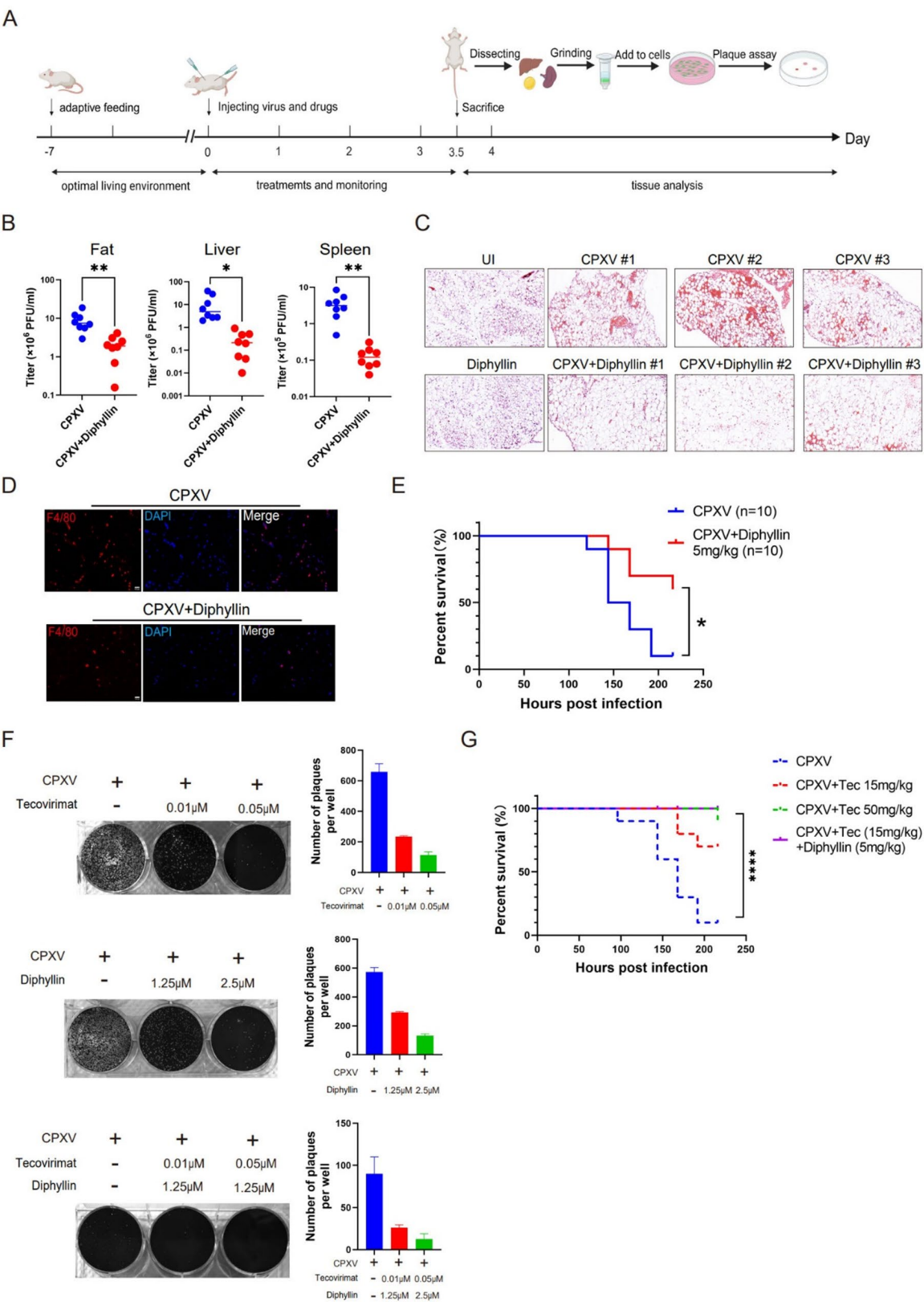


Fig. 5 TMEM175-mediated endolysosomal alkalization restricts poxvirus replication. **(A)** A schematic diagram of the regulation pattern of endolysosomal acidification environment. **(B)** GFP fluorescence imaging of CPXV (MOI=0.05)-infected HeLa cells overexpressing TMEM175 compared with control cells. Plaque assay of supernatant from CPXV-infected HeLa cells overexpressing TMEM175 and control cells. **(C)** Plaque assay and GFP fluorescence imaging of CPXV (MOI=0.05)-infected Vero cells treated with DCPIB at 2.5 μ M and 5 μ M, compared with untreated control. **(D)** Plaque assay and GFP fluorescence imaging of CPXV (MOI=0.05)-infected Vero cells treated with Arachidonic acid (ArA) at 5 μ M and 10 μ M, compared with untreated control. **(E)** Quantitative analysis of E3L (left), A3L (middle), and D13L (right) mRNA expression levels in VACV-infected cells treated or untreated with DCPIB. mRNA levels were measured at 4, 8, 12, and 24 h post-infection by qRT-PCR. **(F)** Western blotting analysis of VACV protein expression in infected cells treated with DCPIB at 5 μ M and 10 μ M or left untreated. Protein levels of E3L, WR117, L1R, and D8L were detected at 4, 8, 12, and 24 h post-infection. The mean value was calculated by one-way ANOVA with Tukey (mean $\pm$ SD, $n = 3$). * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$



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Fig. 6 Diphyllin inhibits poxviral replication in vivo. **(A)** Schematic diagram illustrating the in vivo experimental design and treatment regimen. **(B)** Viral titers measured in various organs (fat, liver, and spleen) collected from mice euthanized after CPXV infection with or without drug treatment. **(C)** H&E staining of fat tissues from mice euthanized after CPXV infection with or without drug treatment. **(D)** Immunofluorescence staining of F4/80 (macrophage marker) and DAPI in adipose tissue from mice euthanized after CPXV infection with or without drug treatment. **(E)** Kaplan–Meier survival curves of mice infected with CPXV and treated with vehicle or Diphyllin (5 mg/kg). Statistical significance was evaluated using the log-rank (Mantel–Cox) test. **(F)** Representative plaque assays and quantification of plaque numbers following treatment with Tecovirimat, Diphyllin, or their combination at the indicated concentrations. Bar graphs represent mean $\pm$ SEM. **(G)** Kaplan–Meier survival analysis of mice receiving CPXV infection followed by treatment with Tecovirimat alone (15 mg/kg or 50 mg/kg), Diphyllin alone (5 mg/kg), or the combination of Diphyllin (5 mg/kg) and low-dose Tecovirimat (15 mg/kg). Statistical significance was evaluated using the log-rank (Mantel–Cox) test

and spleen were harvested for viral titer analysis (Fig. 6A). As shown in Fig. 6B and Fig S6A, Diphyllin treatment significantly reduced viral loads across all examined organs and mitigated weight loss. Consistent with the well-established tropism of orthopoxviruses for adipose tissue, histological analysis of fat tissue revealed that Diphyllin mitigated vascular congestion and inflammatory cell infiltration, indicating an anti-inflammatory effect (Fig. 6C). Immunofluorescence staining further showed a reduction in macrophage presence in adipose tissue (Fig. 6D and S6B). Consistently, cytokine profiling indicated decreased levels of pro-inflammatory cytokines and chemokines, alongside elevated anti-inflammatory mediators (Fig. S6C). To further evaluate the therapeutic relevance of Diphyllin in a clinically meaningful setting, we adopted a lethal CPXV challenge model. Mice treated with Diphyllin (5 mg/kg) exhibited significantly improved survival compared to untreated controls (Fig. 6E), demonstrating that Diphyllin confers protection against lethal poxvirus infection. Inspired by prior studies showing that combining Tecovirimat with other antivirals can produce synergistic effects against orthopoxviruses [15, 16], we tested a Diphyllin–Tecovirimat combination therapy in vivo. This synergy can be mechanistically explained by the fact that Diphyllin targets a host endolysosomal acidification pathway, whereas Tecovirimat blocks the viral protein VP37 [39], thereby providing a rational, complementary strategy to enhance antiviral efficacy and mitigate resistance. We first confirmed that Tecovirimat (ST-246) exhibited potent antiviral activity in vitro, with an EC_{50} of 0.01 μ M and no detectable cytotoxicity up to 200 μ M (Fig. S6D). In plaque reduction assays, both compounds inhibited CPXV replication in a dose-dependent manner. Remarkably, co-treatment with Diphyllin (1.25–2.5 μ M) and low-dose Tecovirimat (0.01–0.05 μ M) resulted in enhanced viral suppression relative to monotherapies (Fig. 6F). Importantly, this combination strategy allowed the use of Tecovirimat at doses significantly lower than those typically required for in vivo efficacy, potentially reducing toxicity risk and treatment cost. In the lethal challenge model, combined Diphyllin and low-dose Tecovirimat provided greater survival protection than either agent alone (Fig. 6G), highlighting the potential of host-virus dual targeting. Taken together, these findings demonstrate that Diphyllin not only limits

viral replication, inflammation, and mortality in vivo but also synergizes with established antivirals, supporting its promise as a host-directed therapeutic strategy against poxvirus infection.

Discussion

In this study, we screened a library of 268 antiviral small-molecule compounds derived from traditional Chinese medicine. Among them, Diphyllin exhibited the most potent inhibitory effect against poxvirus infection and was therefore selected for detailed experimental investigation. In addition to Diphyllin, all other compounds were also evaluated for their antiviral potential to provide a broader reference for our research and to facilitate future drug discovery efforts targeting poxvirus inhibition. Interestingly, Lycorine demonstrated the second-highest inhibition rate of viral infection in our cell-based assays. Previous studies have reported its antiviral activity against Zika virus [40], Chikungunya virus [41], and Dengue virus [42], supporting its potential as a broad-spectrum antiviral agent. Furthermore, compounds such as Arteannuin B, Pentagalloylglucose, Punicalin, Chrysoeriol, Chebulagic acid, 9-Aminoacridine, Lapachol, Decanedioic acid, Dacarbazine, Fipronil, and Gramine also achieved more than 50% inhibition of cowpox virus (CPXV) replication. Notably, Gramine, Arteannuin B, and Pentagalloylglucose have been previously reported to inhibit viruses including Enterovirus71 [43], SARS-CoV-2 [44], and Zika virus [45]. A complete summary of the inhibitory rates of all tested compounds against CPXV is provided in the Supplementary Table 2.

Diphyllin, first reported in 2007 as a novel V-ATPase inhibitor derived from traditional Chinese medicine, was initially explored for its role in suppressing osteoclast-mediated bone resorption [46]. Its identification was based on acridine orange acidification assays, though its precise molecular target had not been defined. Our study is the first to use molecular docking simulations to identify high-affinity binding interactions between Diphyllin and the V_0 and V_1 domain of the V-ATPase, providing a mechanistic framework that solidifies its classification as a *bona fide* V-ATPase inhibitor. While V-ATPase inhibitors have demonstrated antiviral activity against influenza virus [28, 47], Zika [27], coronaviruses [48–50], Ebola [51], and HIV [52], this is the first report of their

efficacy against poxviruses. Compared to other V-ATPase inhibitors, Diphyllin offers multiple advantages. The well characterized inhibitor BafA1 is limited by high systemic toxicity [53, 54], evidenced by liver and spleen damage at doses as low as 350 ng/kg in vivo and by undesired antibiotic properties [55], which render it unsuitable for therapeutic development despite its potent in vitro activity [56]. Meanwhile, KM91104, a more recently developed inhibitor targeting the V-ATPase A3-b2 subunit [57], lacks extensive characterization and shows no superior efficacy to Diphyllin. In contrast, Diphyllin combines broad-spectrum antiviral activity with low cytotoxicity and a well-established safety profile. Although the parent Diphyllin molecule has a modest selectivity index (SI $\approx$ 10), limiting its direct clinical translation, its scaffolds possess substantial medicinal optimization potential. Derivatives have achieved markedly improved potency and SI values, highlighting a promising path for future drug development. For instance, Shen et al. demonstrated that mice treated with 20 mg/kg Diphyllin daily for ten days showed no significant toxicity [58], and Alicia et al. found that a 0.2 mg/kg dose of 6-Deoxyglucose (DGP, a Diphyllin derivative) significantly prolonged survival in Zika virus-infected mice without overt side effects [27]. Consistent with these findings, our study demonstrated that Diphyllin reduced viral burden in CPXV-infected mice and ameliorated inflammatory responses. Given these attributes, Diphyllin emerges as a promising antiviral candidate, with potential for use either as a standalone therapy or in combination with other antiviral agents.

Mechanistically, Diphyllin inhibits vacuolar-type H⁺-ATPase (V-ATPase), thereby impeding proton translocation into the endolysosome. The resulting alkalization interferes with a post-entry stage of the viral life cycle [59, 60]. This mechanism is supported by our complementary finding that activation of the endolysosomal H⁺ efflux channel TMEM175 similarly impairs poxvirus infection (Fig. 7). While our assays employed general acidic vesicle probes (LysoTracker and LysoSensor) that cannot distinguish between late endosomes and lysosomes. Our data are consistent with the established model where poxvirus fusion occurs mainly within late endosomes [14], underscoring the critical role of endolysosomal acidification. To define the entry step targeted, we examined its effects on the viral entry machinery. Diphyllin markedly reduced levels of L1R, a protein essential for the alternative plasma membrane fusion pathway [33]. Furthermore, its potent activity when applied post-binding, contrasted with its inactivity during adsorption, confirms that Diphyllin specifically disrupts the endocytic entry route by acting on post-attachment events.

The intracellular replication mechanisms elucidated in this study using CPXV may have broader implications for

other orthopoxviruses such as MPXV and variola virus. Given the high degree of genomic and structural conservation among orthopoxviruses, including shared dependence on endolysosomal acidification for viral entry, uncoating, and early transcription, it is reasonable to hypothesize that Diphyllin or other V-ATPase inhibitors would exert similar inhibitory effects on MPXV and variola virus replication. Therefore, our findings using CPXV not only delineate a host-targeted antiviral mechanism but also provide a conceptual framework for extending this strategy to combat emerging and re-emerging orthopoxvirus infections in humans.

Our in vivo data further demonstrated that Diphyllin improved the survival of CPXV-infected mice. Importantly, while Diphyllin monotherapy modestly prolonged survival, combining Diphyllin with Tecovirimat resulted in a substantially greater survival benefit. The combination allowed for reduced dosing of each agent while preserving efficacy, suggesting that Diphyllin and Tecovirimat act through complementary mechanisms and may be leveraged in a dose-sparing therapeutic strategy. Recent studies have shown that macrophages are not only key innate immune responders but also a primary in vivo target of orthopoxviruses. Distinct macrophage subsets can be productively infected by VACV, support intracellular viral replication, and generate extracellular enveloped virions (EEV) that facilitate systemic dissemination [61]. Given that adipose tissue contains abundant tissue-resident macrophages with crucial immunometabolic functions, alterations in macrophage-mediated responses are likely to have a direct impact on disease progression during CPXV infection. To explore the underlying mechanism, we analyzed gene expression profiles in adipose tissues, an important peripheral immune-metabolic organ during viral infection. Diphyllin treatment markedly reduced the transcription of several proinflammatory cytokines and chemokines, including Ccl2, Cxcl2, Cxcl1, IL6, and Ccl4, suggesting that Diphyllin effectively dampens the local inflammatory response and limits leukocyte infiltration in adipose tissue. Meanwhile, the increased expression of Arg1 and Ptgs1 indicates a shift toward an anti-inflammatory and tissue-repair phenotype, possibly through M2-like macrophage polarization. The reduced C3 expression further supports the idea that Diphyllin attenuates complement-mediated inflammatory injury. Taken together, these results suggest that Diphyllin exerts a systemic protective effect not only by inhibiting viral replication but also by reprogramming immune responses. By alleviating inflammation and promoting an anti-inflammatory milieu, Diphyllin improves host tolerance to infection, thereby prolonging survival in CPXV-infected mice.

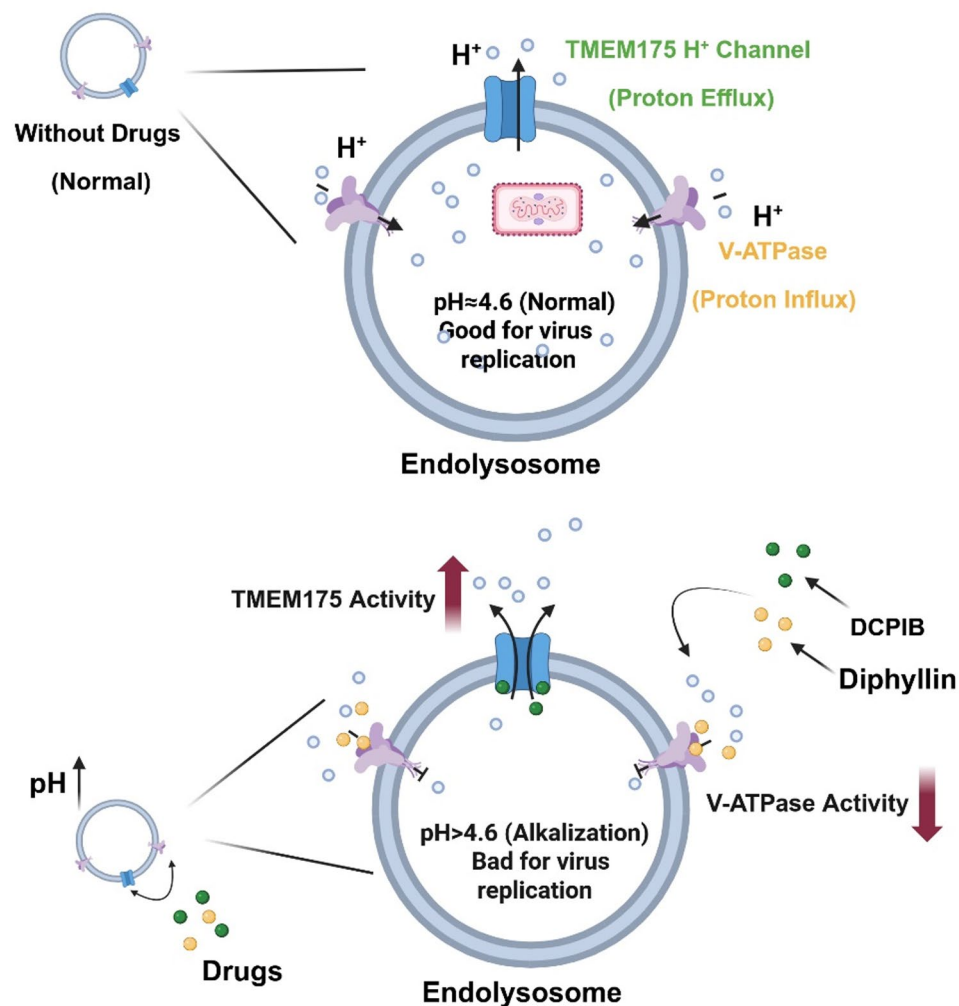


Fig. 7 TMEM175 and V-ATPase coordinate endolysosomal pH regulation to control poxvirus replication. Under physiological conditions, the coordinated activity of TMEM175 (facilitating H⁺ efflux) and V-ATPase (driving H⁺ influx) maintains endolysosomal pH at approximately 4.6. This acidic environment is critical for initiating viral membrane fusion and subsequent genome release into the cytosol. In contrast, pharmacological activation of TMEM175 (e.g., with DCPIB) or inhibition of V-ATPase (e.g., with Diphyllin) disrupts this equilibrium, raises endolysosomal pH (> 4.6), and ultimately impairs viral replication

Conclusions

In conclusion, our study provides strong evidence that endolysosomal acidification is essential for poxvirus entry and replication. Using multiple lines of evidence, including L1R suppression, LysoTracker acidification assays, low pH rescue experiments, and mouse survival studies, we consistently found that blocking endolysosomal acidification significantly impairs poxvirus infection. These combined in vitro and in vivo findings highlight the critical role of endolysosomal pH in the poxvirus life cycle and suggest that targeting this pathway represent an effective antiviral strategy against poxviruses (Fig. 7).

Abbreviations

CPXV	Cowpox virus
VACV	Vaccinia virus
MPXV	Monkeypox virus
EC ₅₀	Half-maximal effective concentration

V-ATPase	Vacuolar H ⁺ -ATPase (proton pump)
TEM	Transmission electron microscopy
qPCR	Quantitative polymerase chain reaction
ArA	Arachidonic acid
PFU	Plaque-forming units
MOI	Multiplicity of infection
siRNA	Small interfering RNA
IMV	Intracellular mature virion
ipTM	Interface predicted TM-score
AF3	AlphaFold 3
PLIP	Protein-Ligand Interaction Profiler (binding analysis tool)

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12964-026-02705-6>.

- Supplementary Material 1.
- Supplementary Material 2.
- Supplementary Material 3.
- Supplementary Material 4.

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The authors explicitly declare that the entire research process was carried out in the absence of any commercial or financial relationships which could potentially be interpreted as a conflict of interest.

Authors' contributions

S. Xu, X. Zhang and C. Li contribute equally to this work. Z. Liu and F. Chan conceptualized and designed the project. Y. He and Y. Lv contributed to manuscript review and editing. Z. Liu, S. Xu, and X. Zhang performed the experiments, analyzed the data, and drafted the initial manuscript. C. Li assisted with experimental procedures and data analysis. X. Wang and C. Wan were responsible for histological section evaluation. Z. Zhang and P. Wang contributed to sample collection and data organization. All authors made substantial contributions to the work and approved the final version of the manuscript.

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Data availability

The original contributions presented within this study are incorporated in the article itself or in the Supplementary Material. For any further inquiries regarding the data, please contact the corresponding authors.

Declarations

Ethics approval and consent to participate

This study underwent review and received approval from the First Affiliated Hospital of Xi'an Jiaotong University (approval number: XJTUAE2024619).

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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References

- Moss B. Membrane fusion during poxvirus entry. *Semin Cell Dev Biol.* 2016;60:89–96.
- Greseth MD, Traktman P. The life cycle of the vaccinia virus genome. *Annual Rev Virol.* 2022;9(1):239–59.
- Moss B. Poxvirus DNA replication. *Cold Spring Harb Perspect Biol.* 2013;5(9):a010199.
- Sherwat A, et al. Tecovirimat and the treatment of monkeypox - past, present, and future considerations. *N Engl J Med.* 2022;387(7):579–81.
- Rizk JG, et al. prevention and treatment of monkeypox. *Drugs.* 2022;82(9):957–63.
- O'Shea J, et al. Interim guidance for prevention and treatment of monkeypox in persons with HIV infection - United States, August 2022. *MMWR Morb Mortal Wkly Rep.* 2022;71(32):1023–8.
- Desai AN, et al. Compassionate use of tecovirimat for the treatment of monkeypox infection. *JAMA.* 2022;328(13):1348–50.
- Ali R, et al. Tecovirimat for clade I MPXV infection in the Democratic Republic of Congo. *N Engl J Med.* 2025;392(15):1484–96.
- Staff GB. Tecovirimat shown to lack efficacy as treatment for clade II mpxv. 2025. Available from: <https://globalbiodefense.com/2025/03/12/tecovirimat-shown-to-lack-efficacy-as-treatment-for-clade-ii-mpox/>. Cited 2026 January 12.
- Andreï G, Snoeck R. Cidofovir activity against poxvirus infections. *Viruses.* 2010;2(12):2803–30.
- Foster SA, Parker S, Lanier R. The role of brincidofovir in preparation for a potential smallpox outbreak. *Viruses.* 2017;9(11):320.
- Lanier R, et al. Development of CMX001 for the treatment of poxvirus infections. *Viruses.* 2010;2(12):2740–62.
- Hutson CL, et al. Pharmacokinetics and efficacy of a potential smallpox therapeutic, brincidofovir, in a lethal monkeypox virus animal model. *MSphere.* 2021;6(1):e00126-21.
- Yang G, et al. An orally bioavailable antipoxvirus compound (ST-246) inhibits extracellular virus formation and protects mice from lethal orthopoxvirus challenge. *J Virol.* 2005;79(20):13139–49.
- Witwit H, et al. Repurposing drugs for synergistic combination therapies to counteract monkeypox virus tecovirimat resistance. *Viruses.* 2025;17(1):92.
- Siegrist EA, Sassine J. Antivirals with activity against mpxv: a clinically oriented review. *Clin Infect Dis.* 2023;76(1):155–64.
- Townsend AC, et al. Vaccinia virus entry into cells via a low-pH-dependent endosomal pathway. *J Virol.* 2006;80(18):8899–908.
- White JM, Whittaker GR. Fusion of enveloped viruses in endosomes. *Traffic.* 2016;17(6):593–614.
- Townsend AC, Moss B. Two distinct low-pH steps promote entry of vaccinia virus. *J Virol.* 2007;81(16):8613–20.
- Pamathy S, et al. The curious case of vacuolar atpase: regulation of signaling pathways. *Mol Cancer.* 2018;17(1):41.
- Song Q, et al. The emerging roles of vacuolar-type ATPase-dependent lysosomal acidification in neurodegenerative diseases. *Translational Neurodegeneration.* 2020;9(1):17.
- Zhang Z, et al. Synthesis and identification of cytotoxic diphyllin glycosides as vacuolar H(+)-ATPase inhibitors. *Eur J Med Chem.* 2014;82:466–71.
- Cui Q, et al. Lignans and their derivatives from plants as antivirals. *Molecules.* 2020;25(1):183.
- Lv J-P, et al. New cyclopeptide alkaloids from the whole plant of *Justicia procumbens* L. *Nat Prod Res.* 2021;35(21):4032–40.
- Ramesh C, et al. Arylnaphthalide lignans from *Cleistanthus collinus*. *Chem Pharm Bull.* 2003;51(11):1299–300.
- Prieto JM, et al. Topical anti-inflammatory lignans from *Haplophyllum hispanicum*. *Z Fur Naturforschung C J Biosci.* 1996;51(9–10):618–22.
- Martinez-Lopez A, et al. Glycosylated diphyllin as a broad-spectrum antiviral agent against Zika virus. *EBioMedicine.* 2019;47:269–83.
- Chen H-W, et al. Inhibitory and combinatorial effect of diphyllin, a V-ATPase blocker, on influenza viruses. *Antiviral Res.* 2013;99(3):371–82.
- Štefánik M, et al. Diphyllin shows a broad-spectrum antiviral activity against multiple medically important enveloped RNA and DNA viruses. *Viruses.* 2022;14(2):354.
- Xu H, Ren D. Lysosomal physiology. *Annu Rev Physiol.* 2015;77:57–80.
- Hu M, et al. Parkinson's disease-risk protein TMEM175 is a proton-activated proton channel in lysosomes. *Cell.* 2022;185(13):2292–2308.e20.
- Abramson J, et al. Accurate structure prediction of biomolecular interactions with alphafold 3. *Nature.* 2024;630(8016):493–500.
- Hou F, et al. mRNA vaccines encoding fusion proteins of monkeypox virus antigens protect mice from vaccinia virus challenge. *Nat Commun.* 2023;14(1):5925.
- Salvi A, et al. PHY34 inhibits autophagy through V-ATPase V0A2 subunit inhibition and CAS/CSE1L nuclear cargo trafficking in high grade serous ovarian cancer. *Cell Death Dis.* 2022;13(1):45.
- Forgac M. Vacuolar atpases: rotary proton pumps in physiology and pathophysiology. *Nat Rev Mol Cell Biol.* 2007;8(11):917–29.
- Srinivasan S, et al. Crystal structure of the cytoplasmic N-terminal domain of subunit I, a homolog of subunit a, of V-ATPase. *J Mol Biol.* 2011;412(1):14–21.

37. Stransky L, Cotter K, Forgac M. The function of V-ATPases in cancer. *Physiol Rev*. 2016;96(3):1071–91.
38. Post S, Rueschpler L, Schloer S. Endolysosomes as a sorting hub for emerging viruses: gatekeepers of cellular defense, viral fate and promising therapeutic target. *Pharmacol Res*. 2025;222:108051.
39. Gigante CM, et al. Notes from the field: mpox cluster caused by tecovirimat-resistant monkeypox virus - five states, October 2023–February 2024. *MMWR Morb Mortal Wkly Rep*. 2024;73(40):903–5.
40. Chen H, et al. Antiviral activity of Lycorine against Zika virus in vivo and in vitro. *Virology*. 2020;546:88–97.
41. Rabelo VW, et al. Antiviral evaluation of 1,4-disubstituted-1,2,3-triazole derivatives against Chikungunya virus. *Future Virol*. 2023;18(13):865–80.
42. Wang P, et al. Anti-dengue-virus activity and structure-activity relationship studies of lycorine derivatives. *ChemMedChem*. 2014;9(7):1522–33.
43. Wei Y, et al. Discovery of Gramine derivatives that inhibit the early stage of EV71 replication in vitro. *Molecules*. 2014;19(7):8949–64.
44. Varela K, et al. Inhibition of cysteine proteases via thiol-Michael addition explains the Anti-SARS-CoV-2 and bioactive properties of Arteannuin B. *J Nat Prod*. 2023;86(7):1654–66.
45. Sharma N, Kumar P, Giri R. Polysaccharides like pentagalloylglucose, parishin A and stevioside inhibits the viral entry by binding the Zika virus envelope protein. *J Biomol Struct Dyn*. 2021;39(16):6008–20.
46. Sørensen MG, et al. Diphyllin, a novel and naturally potent V-ATPase inhibitor, abrogates acidification of the osteoclastic resorption lacunae and bone resorption. *J Bone Mineral Research: Official J Am Soc Bone Mineral Res*. 2007;22(10):1640–8.
47. Müller KH, et al. The proton translocation domain of cellular vacuolar ATPase provides a target for the treatment of influenza A virus infections. *Br J Pharmacol*. 2011;164(2):344–57.
48. Hu CJ, et al. Nanoparticulate vacuolar ATPase blocker exhibits potent host-targeted antiviral activity against feline coronavirus. *Sci Rep*. 2017;7(1):13043.
49. Stefanik M, et al. Antiviral activity of vacuolar ATPase blocker diphyllin against SARS-CoV-2. *Microorganisms*. 2021;9(3):471.
50. Icho S, et al. Dual Inhibition of Vacuolar-ATPase and TMPRSS2 is required for complete Blockade of SARS-CoV-2 entry into cells. *Antimicrob Agents Chemother*. 2022;66(7):e0043922.
51. Plescia CB, et al. Evaluation of phenol-substituted diphyllin derivatives as selective antagonists for ebola virus entry. *ACS Infect Dis*. 2022;8(5):942–57.
52. Zhang HJ, et al. Anti-HIV diphyllin glycosides from *Justicia Gendarussa*. *Phytochemistry*. 2017;136:94–100.
53. Dröse S, Altendorf K. Bafilomycins and concanamycins as inhibitors of V-ATPases and P-ATPases. *J Exp Biol*. 1997;200(Pt 1):1–8.
54. Mauvezin C, Neufeld TP. Bafilomycin A1 disrupts autophagic flux by inhibiting both V-ATPase-dependent acidification and Ca-P60A/SERCA-dependent autophagosome-lysosome fusion. *Autophagy*. 2015;11(8):1437–8.
55. Fiore AE, et al. Antiviral agents for the treatment and chemoprophylaxis of influenza --- recommendations of the advisory committee on immunization practices (ACIP). *MMWR Recomm Rep*. 2011;60(1):1–24.
56. Wang R, et al. Molecular basis of V-ATPase inhibition by Bafilomycin A1. *Nat Commun*. 2021;12(1):1782.
57. Marrone G, et al. The adenosine monophosphate-activated protein kinase-vacuolar adenosine triphosphatase-pH axis: a key regulator of the profibrogenic phenotype of human hepatic stellate cells. *Hepatology*. 2018;68(3):1140–53.
58. Shen W, et al. Effects of diphyllin as a novel V-ATPase inhibitor on gastric adenocarcinoma. *Eur J Pharmacol*. 2011;667(1–3):330–8.
59. Bruneau RC, Tazi L, Rothenburg S. Cowpox viruses: a zoo full of viral diversity and lurking threats. *Biomolecules*. 2023;13(2):325.
60. Talarek E, Marczyńska M. Cowpox virus infection. *N Engl J Med*. 2018;378(2):181.
61. Byrd D, et al. Primary human macrophages serve as vehicles for vaccinia virus replication and dissemination. *J Virol*. 2014;88(12):6819–31.

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